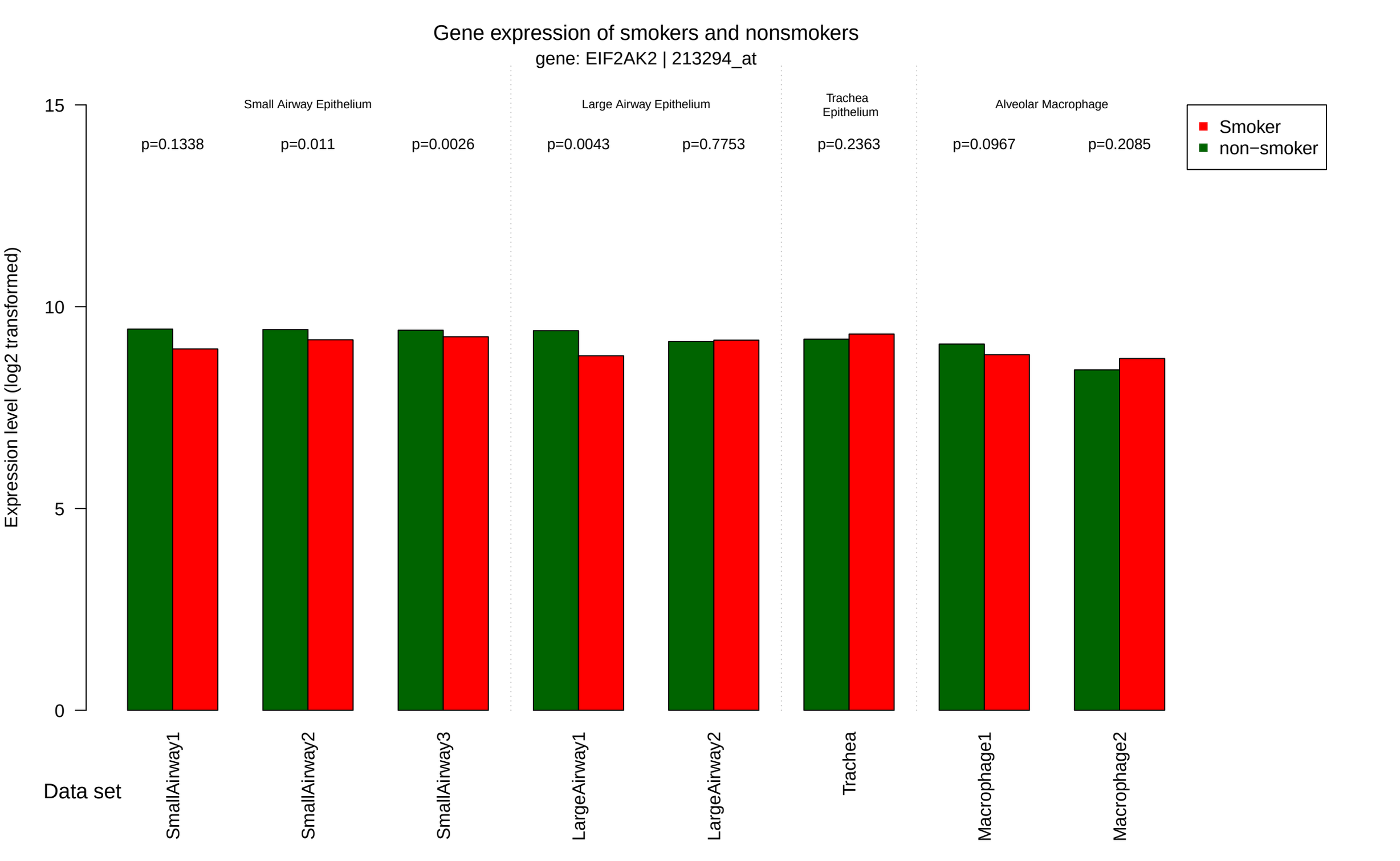
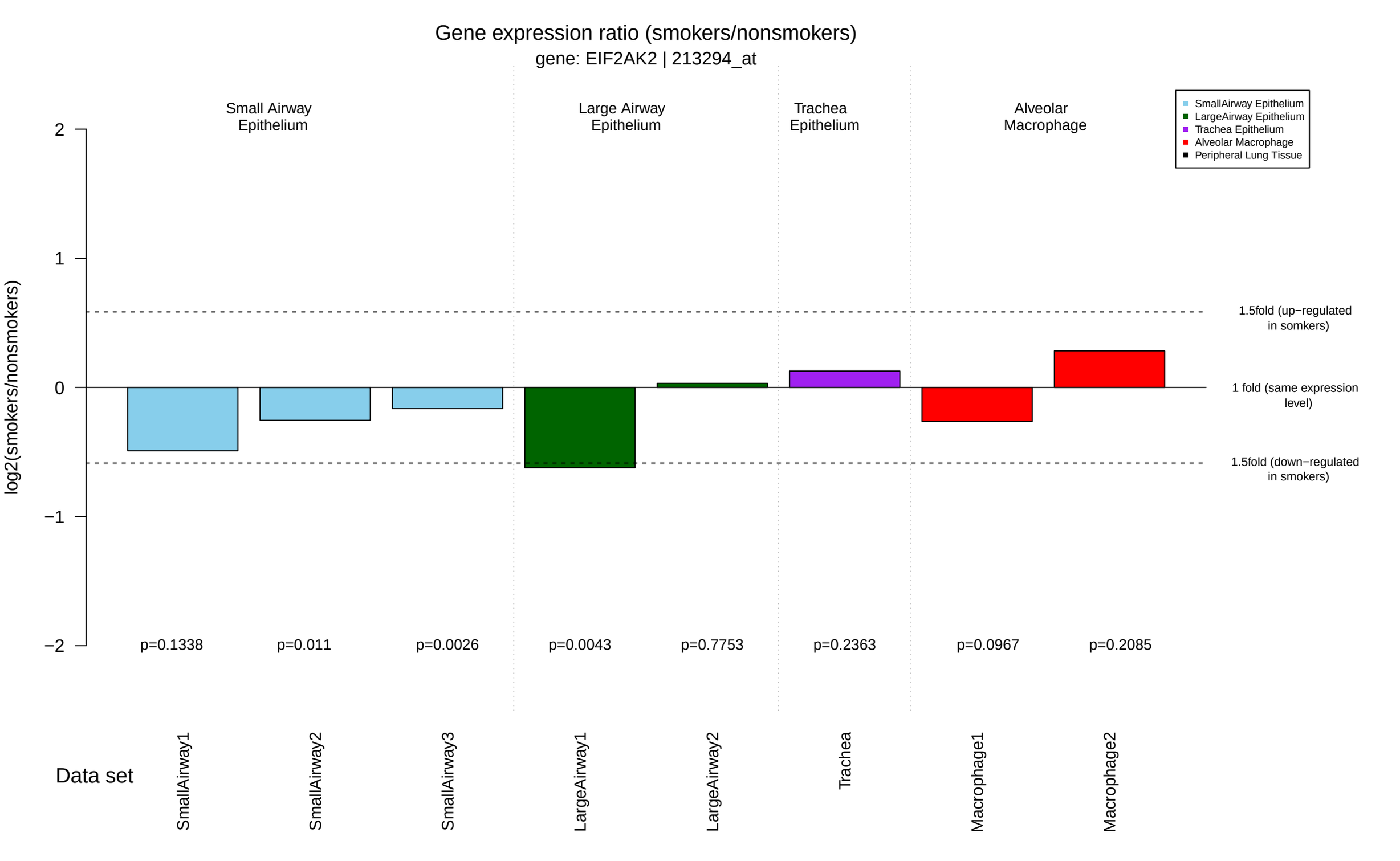
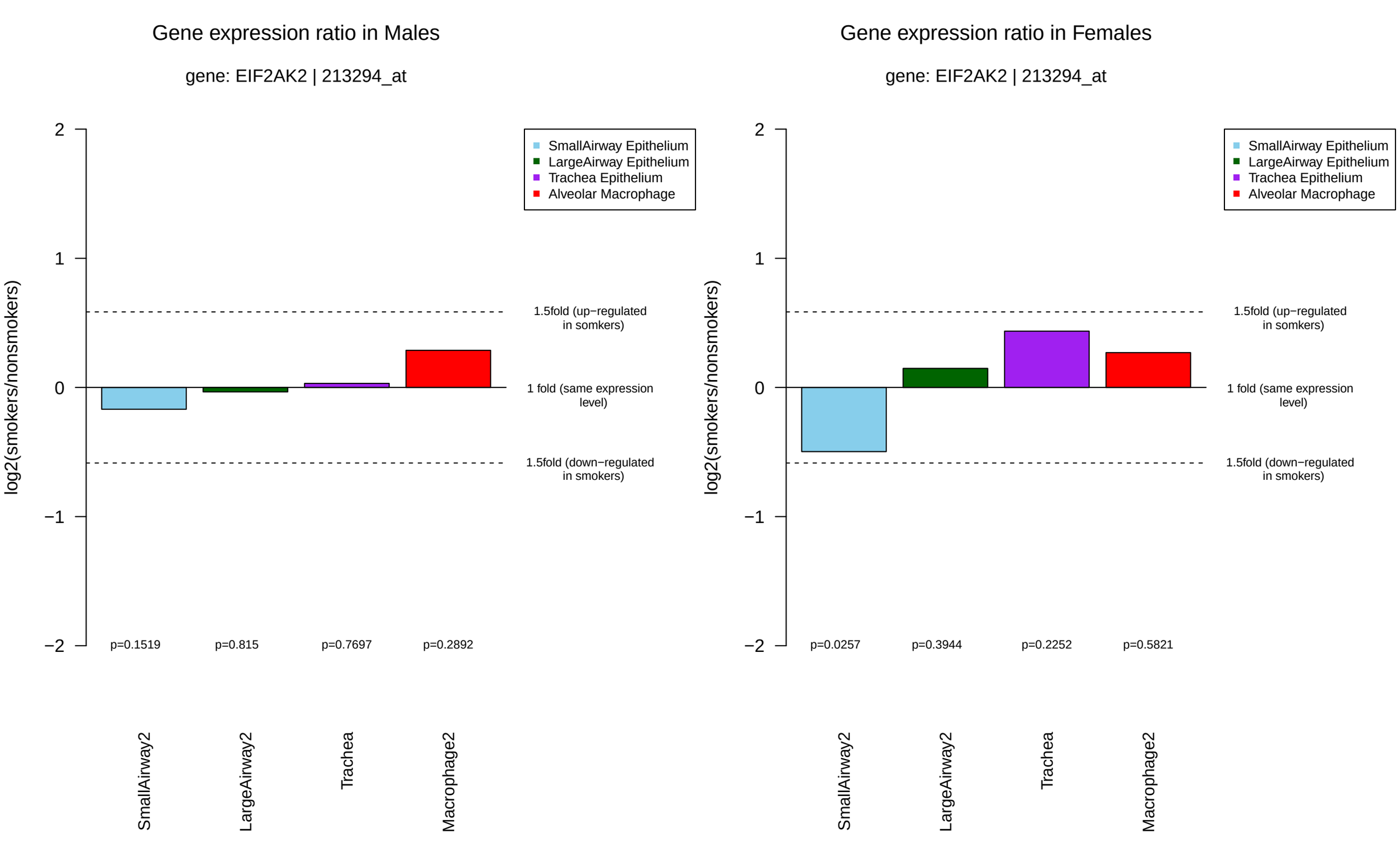
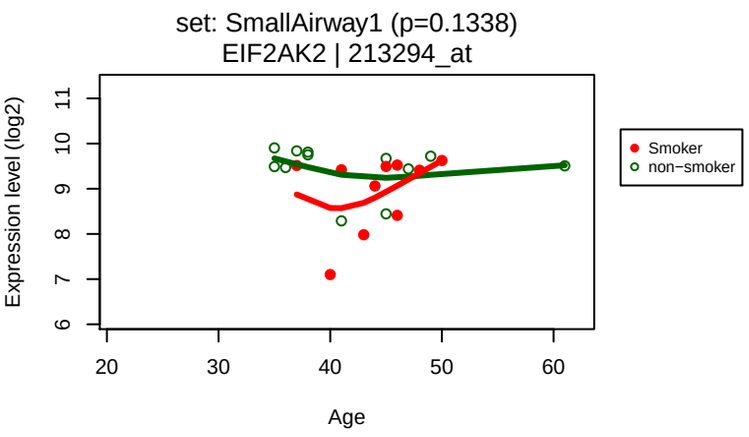
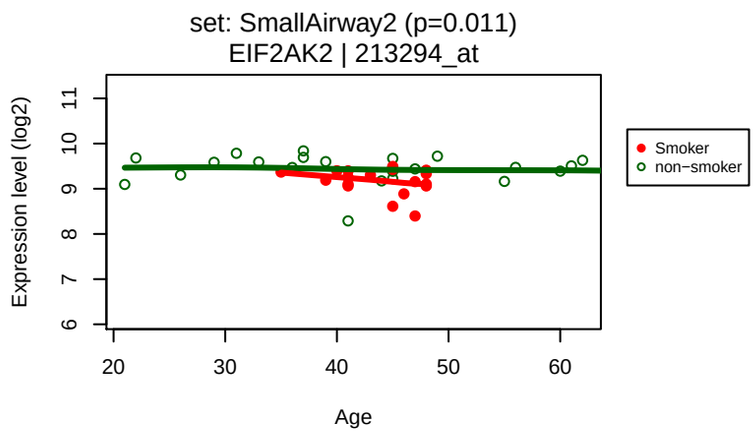
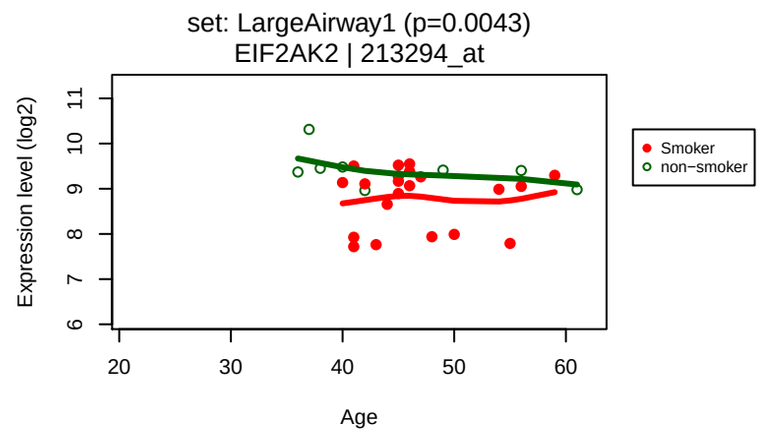

Gene expression of smokers and nonsmokers
gene: EIF2AK2 | 213294_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1338
p=0.011
p=0.0026
p=0.0043
p=0.7753
p=0.2363
p=0.0967
p=0.2085
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

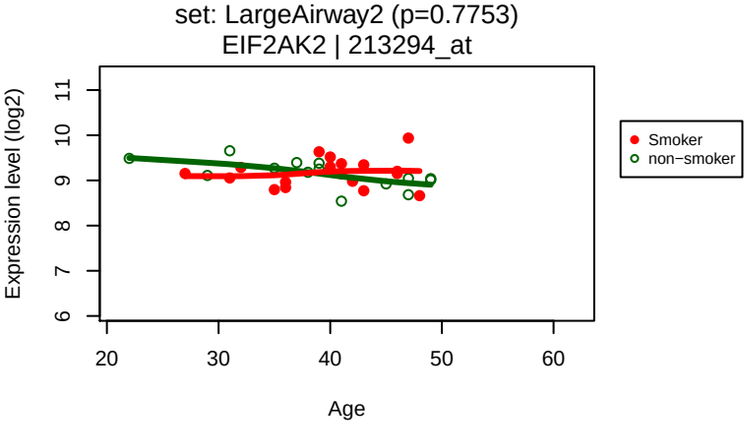

Gene expression ratio (smokers/nonsmokers)
gene: EIF2AK2 | 213294_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1338
p=0.011
p=0.0026
p=0.0043
p=0.7753
p=0.2363
p=0.0967
p=0.2085
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

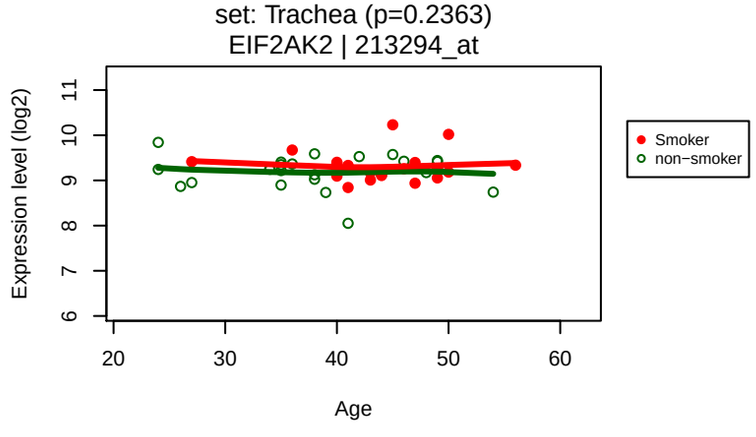

Gene expression ratio in Males
gene: EIF2AK2 | 213294_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1519
p=0.815
p=0.7697
p=0.2892
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EIF2AK2 | 213294_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0257
p=0.3944
p=0.2252
p=0.5821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

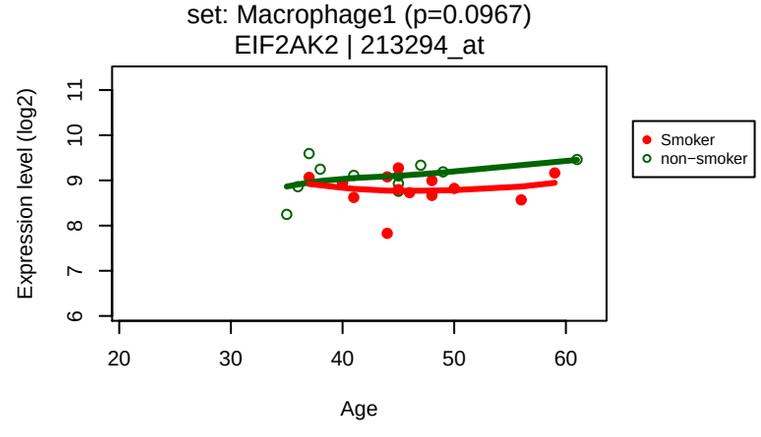

set: SmallAirway1 (p=0.1338)
EIF2AK2 | 213294_at
Expression level (log2)
Age
Smoker
non−smoker

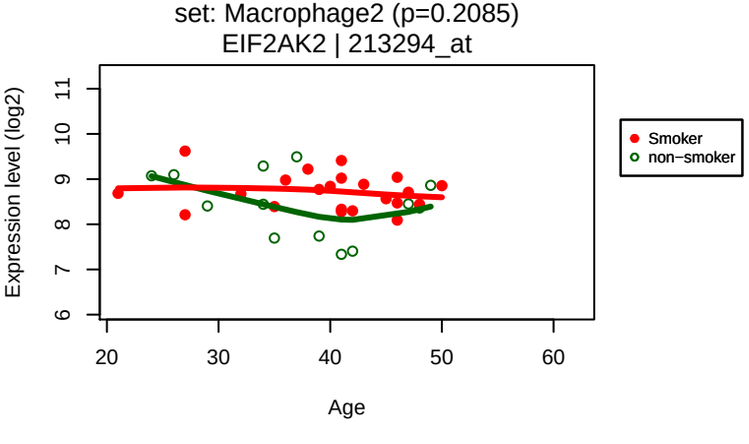

set: SmallAirway2 (p=0.011)
EIF2AK2 | 213294_at
Expression level (log2)
Age
Smoker
non−smoker

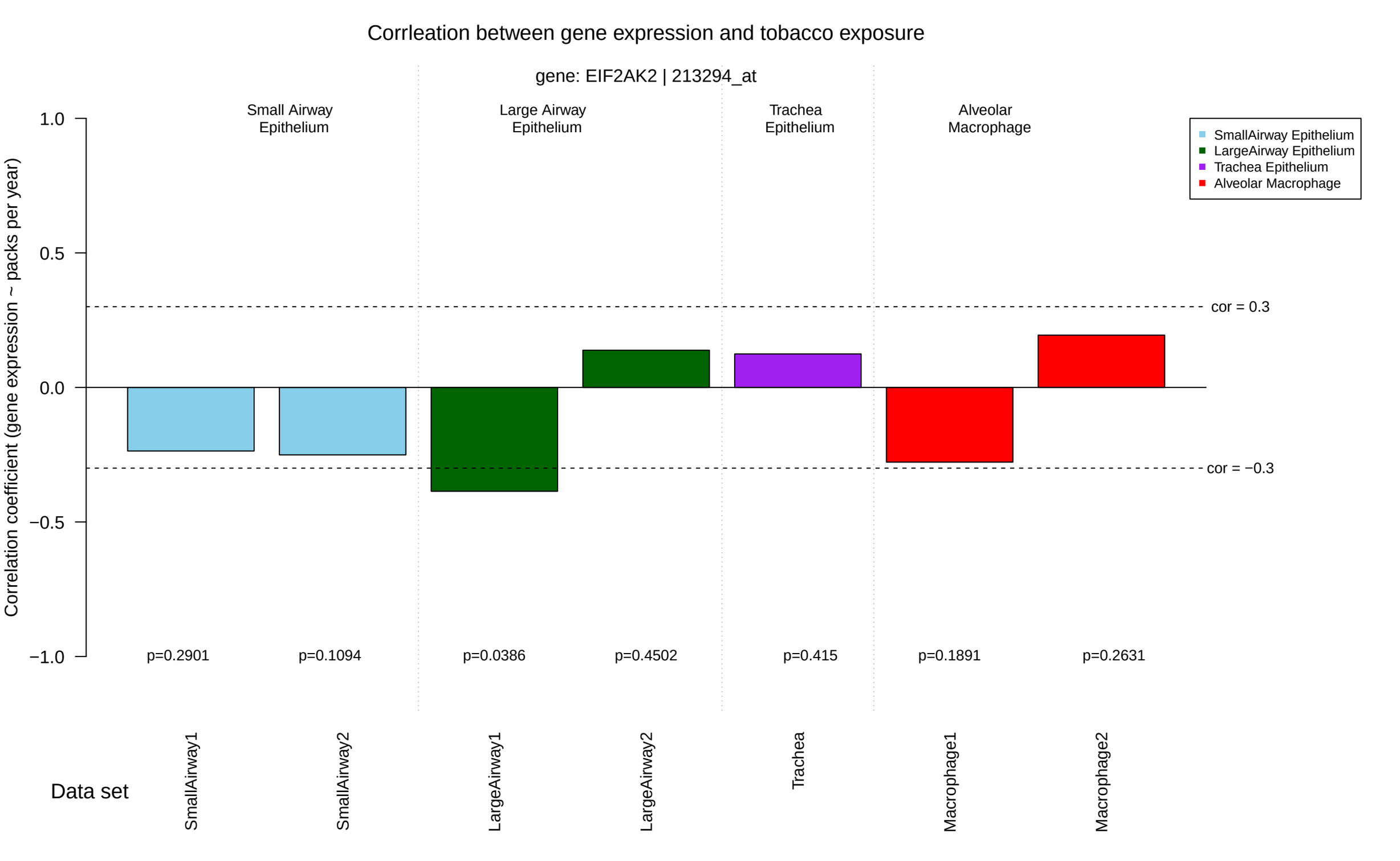

set: LargeAirway1 (p=0.0043)
EIF2AK2 | 213294_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7753)
EIF2AK2 | 213294_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2363)
EIF2AK2 | 213294_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0967)
EIF2AK2 | 213294_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2085)
EIF2AK2 | 213294_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EIF2AK2 | 213294_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2901
p=0.1094
p=0.0386
p=0.4502
p=0.415
p=0.1891
p=0.2631
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2